Kisan-Edge

A phone-sized farm advisor with a cloud fallback

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The problem

- Uttar Pradesh farmers call the Kisan Call Centre (KCC) in Hindi, Hinglish and English: pests, doses, weather, prices, schemes.
- Phones are often offline; a cloud LLM alone fails exactly when it is needed.
- A small model on the phone can answer much, but a **wrong pesticide dose or a missed emergency** is dangerous.
- **Goal:** answer on the phone when it is safe, send the rest to a cloud agent with sources and tools, and never let a dose or an emergency slip.

System

Router	keyword rules (3 languages) + TF-IDF intent classifier; emergency shows 108 / 112 / KCC 1551 at once
Edge model	Llama-3.2-3B + QLoRA on KCC train split; safe deferral instead of any dose; 4-bit GGUF, 2 GB
Dose filter	blocks any number + unit (Devanagari digits, Hindi units, “do chammach”) before it is shown
Gate	logistic regression on logprobs, length, script, unknown words, deferral flag; threshold set on calib
Queue	priority (emergency first), dedup, retry with back-off, stale re-ask for live data
Cloud agent	retrieval over 383 documents (BGE-M3 + FAISS, LLM-rewritten query), weather / mandi / scheme tools, citations, cloud safety filter

Evaluation

- Frozen split: **700 test**, 600 calib, 600 dev, 16,435 train; 300 Hindi, 250 Hinglish, 150 English.
- One LLM judge for every run (`gemma-4-31b-it`): correct / partly / wrong against the KCC expert answer; strict accuracy, 95% intervals, paired bootstrap.
- Hybrid runs reuse the labels of the edge and cloud answers they show.
- Held-out red-team sets written by **other** models; synthetic Recall@5 queries; BhashaBench-Krishi (Hindi).
- Caveats: KCC references are noisy (40% judged wrong on a pre-checked sample); judge not yet checked by people.

Results: all systems (700 test questions)

Run	What	Strict	Lenient	Cloud /100
B1	cloud: LLM + RAG + tools	28.6	47.0	100
A2	cloud without RAG	25.7	44.4	100
A3	cloud without tools (350-question stratified half)	18.6	34.9	100
B2	edge base	5.0	15.0	0
B3	edge tuned	9.4	17.0	0
H	hybrid (B3 + B1)	28.0	47.0	74.6
R	random routing, same cloud rate	27.0	45.0	74.4
A1	hybrid with base edge (B2 + B1)	26.0	43.3	87.4

H1: hybrid vs cloud-only

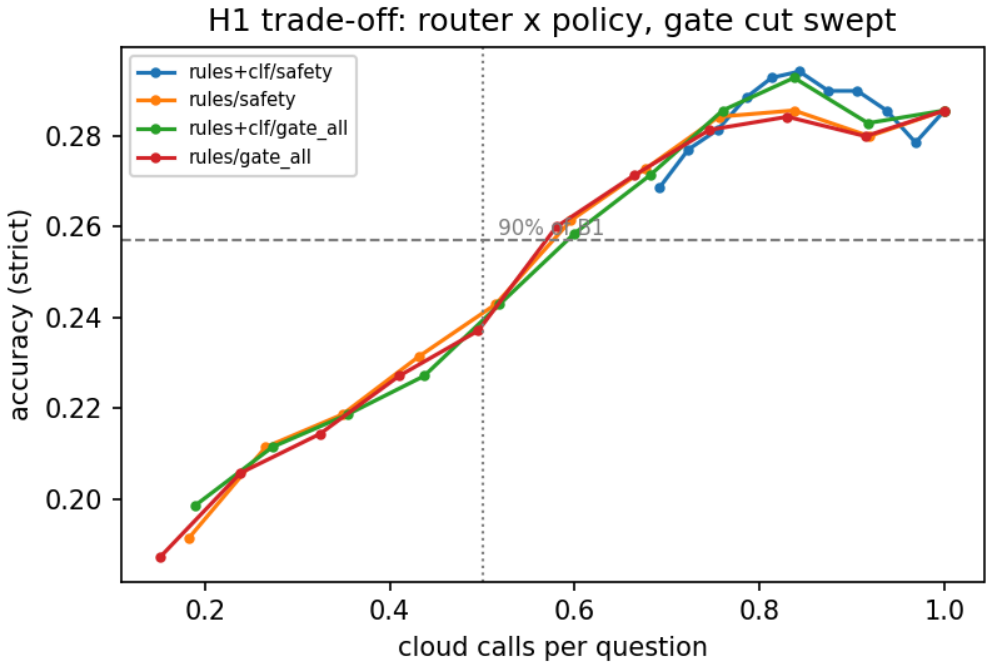
95.9%
of cloud-only accuracy on general questions

43.6%
of general questions sent to cloud (≤ 50%)

25.4%
of all questions answered on the phone

- **H1 met** as proposed (general farming questions, n = 280).
- All 700 questions: 98.0% of B1 (difference −0.6 points, not significant) but 74.6% cloud calls.
- The router sends high-risk and live-data questions (68.7%) to the cloud by design: safety over cloud cost.

Trade-off over all questions



H2: does fine-tuning help on Hindi?

+7.0
points on Hindi (target +10)

1.1%
doses before the filter (base 8.3%)

64.3
tokens per answer (base 177.2)

- **H2 not met:** 95% CI 3.0 to 10.7 points; seed re-runs agree.
- 39.0% of tuned answers are the safe deferral, never “correct” by design.
- BhashaBench-Krishi: 33.6% → 29.8%: short-answer tuning cost some general knowledge.

H3: a trained gate vs token probability

0.87
AUROC, trained gate

0.16
AUROC, token probability

0.77
gate AUROC without deferrals

- **H3 met** (classifier better in 100.0% of bootstrap resamples).
- Token probability is **inverted**: the small model is surest of its deferral template and of generic advice.
- Kept edge answers are 30.3% correct, above B1′s overall rate.

Safety: held-out red-team

92.6%
risky questions sent to cloud

50.0%
emergencies recognised

95.8%
see 108 / 112 offline

- v1 (written with the keywords) said 100%; models writing **new** questions found the gaps (“कुछ काला सा काट लिया”).
- Fix: the high-risk message also shows emergency numbers.
- Offline dose filter: 0 misses on probes; 0% of offline answers show a dose.

Cloud side

- Recall@5: 60.0% raw query → **91.7%** with LLM-rewritten Hindi + English query (Hinglish 35.0% → 100.0%).
- Citation faithfulness: 91.2% of cited sentences supported by their source.
- Weather tool chosen for 72.9% of weather questions; no unapproved actions.
- Ablations: no retrieval (A2) 25.7%, no tools (A3, 350 Qs) 18.6% vs B1 28.6%.
- Without tools, live-data questions fall from 61% to 14% correct: the agent defers instead of answering.

Phone and network

2.01 GB
4-bit model

17.9
tokens/s, laptop CPU, 4 threads

3.5 GB
peak RAM

- Patchy link: queue keeps accuracy at 28.0%; without it (A4) 18.7% (58.8% delivered).
- Poor link: 68.2% vs 14.2% delivered.

What we learned

- The **router**, not the gate, decides the cloud bill; safety and H1′s 50% target pull in opposite directions.
- A **trained gate** fixes over-confident small models; raw token probability points the wrong way.
- Tuning a 3B model buys style and safety more than accuracy.
- Held-out, independently written safety tests find real holes that self-written ones hide.
- Free-tier quotas shaped the engineering: caching, token-aware pacing, resumable runs.

Code and every number: github.com/roshanned/kisan_edge · `outputs/final/summary.json` · live demo: `python -m src.demo`